

E-Commerce Web Program

Comprehensive Study Guide & Notes

Based on Course Materials: "3209 E-commerce Web Program Mid.pdf"

Includes Definitions, Business Models, Technologies, and Architectures

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1. Introduction to E-Commerce & E-Business

Definition of E-Commerce

E-commerce is an abbreviation used for electronic commerce. It is the process through which the buying, selling, dealing, ordering, and paying for goods and services are done over the internet. In this type of online commercial transaction, the seller can communicate with the buyer without having a face-to-face interaction.

Formal Definition: E-commerce is the use of the internet, the web, and mobile platforms to conduct business through digitally enabled commercial transactions between and among organizations and individuals.

Definition of E-Business

Electronic Business, shortly known as E-business, is the online presence of a business. It can also be defined as the business which is done with the help of the internet or electronic data interchange. E-commerce is one of the important components of E-business, but it is not the entirety of it.

E-business is not confined to buying and selling of goods only; it includes other activities that form part of the business, such as providing services to customers, communicating with employees, and collaborating with business partners. All basic business operations are done using electronic media.

Key Differences Between E-Commerce and E-Business

Feature	E-Commerce	E-Business
1. Core Meaning	Buying and selling goods or services online.	Digitally managing all internal and external company operations.
2. Boundary Reached	Reaches outside the company to connect with customers and suppliers.	Stays inside the company to run internal systems and daily work.
3. Exchange of Value	Always involves money or financial transactions.	Does not require a financial exchange or money transfer.
4. Scope	A narrower concept (it is a subset of e-business). Connects with external buyers and sellers directly.	A broader concept (it encompasses e-commerce entirely).
5. Target Audience	External customers, suppliers, and distributors.	Connects internal employees, departments, and operations.
6. Primary Goal	Focused on revenue generation and online sales.	Focused on improving operational efficiency and workflow.
7. Technology Used	Uses the public internet, web, and mobile platforms.	Uses internal networks, intranets, and Enterprise Systems (ERP).

2. Characteristics of Traditional vs. E-Commerce

Why Study E-Commerce?

Technological Shift: E-commerce technology is fundamentally different and significantly more powerful than any commercial technology that preceded it.

Commercial Transformation: It introduces structural changes to how commerce operates globally, altering traditional retail and supply models.

Note on MRO (Maintenance, Repair, and Operations):

MRO refers to supplies consumed during the production process that do not become part of the final product and are not central to the firm's core output. They are commonly categorized as indirect goods. Examples include consumables (cleaning products, office supplies), industrial equipment (pumps, valves), plant upkeep (spare parts, lubricants), and workplace assets (computers, furniture).

Characteristics of Traditional Commerce

To understand why E-commerce is revolutionary, the points below contrast it with the key features that defined traditional commerce:

- **Consumers as Passive Targets:** Consumers are treated as passive recipients of marketing messages and mass products pushed by sellers without much direct interaction or feedback. *Example: Watching a billboard on a highway — anyone absorbs the message as a spectator without any way to click, question, or interact on the spot.*
- **Mass-Marketing Driven:** Marketing relies heavily on broad mass-media broadcasting. *Example: Printing a flyer for a local bakery and distributing it to thousands of households without the information whether residents buy baked goods or not.*
- **Sales-Force Driven:** Operations depend heavily on physical sales staff to interact with customers. *Example: Buying a suit at a physical shop where a salesman talks to the buyer in person and convinces him to buy that product.*
- **Fixed Prices:** Prices are predetermined by merchants with limited dynamic adjustments. *Example: Buying shoes in a shop with a 500 BDT price tag that the buyer cannot bargain or negotiate lower.*
- **Information Asymmetry:** Merchants possess significantly more information about product quality, true cost, and availability than consumers do. *Example: Buying a used car where the dealer knows the engine has hidden problems, but the buyer does not.*

3. Eight Unique Features of E-Commerce Technology

E-commerce technologies possess unique characteristics that differentiate them from traditional commercial systems. These eight features define the modern digital marketplace.

1. Ubiquity

Technology is available everywhere, anytime, creating a "marketplace" that replaces physical location constraints. E-Commerce is accessible everywhere at all times. Consumers can browse online stores 24 hours a day from their mobile phones without visiting a physical shop or leaving their home or workplace.

2. Global Reach

Technology reaches across national boundaries to connect with customers. E-Commerce technology breaks down geographic barriers. A merchant can sell goods or services to buyers in different countries just like a local customer.

3. Universal Standards

There is one uniform set of internet standards shared globally across platforms. E-Commerce systems run on a single set of shared technology standards defined by the internet. Because systems worldwide use identical technical rules, different software and payment gateways can seamlessly interact.

4. Information Richness

Detailed, multimedia-rich information combining text, audio, and video is delivered to the user. E-commerce offers an engaging shopping environment by combining images, text descriptions, video demonstrations, and audio. This helps consumers to accurately evaluate the quality of a product.

5. Interactivity

Technology enables two-way communication between buyers and sellers. E-Commerce platforms allow active user engagement. Consumers can search for options, ask questions via live chat or chatbots, and leave product ratings and feedback.

6. Information Density

The total amount and quality of information available to all market participants increases rapidly while costs decline. E-Commerce makes market information vast, accurate, and cheap to access. Buyers can easily perform real-time price comparisons with minimal effort.

7. Personalization and Customization

Messages and products can be adjusted according to individual user preferences and behavior. Online platforms can dynamically adapt marketing content and recommendations to match an individual's personal search history and purchase records. Sellers can also offer customized options to customers.

8. Social Technology

Technology enables user-generated content and social network sharing. E-Commerce encourages consumers to create their own content, such as unboxing videos and photo reviews. Users can share experiences and give ratings as well.

4. Types of E-Commerce

E-Commerce is classified either by the market relationship or by the technology platform used for the transaction.

- **Business-to-Consumer (B2C):** Online businesses sell products, services, or digital content directly to individual buyers. A company opens a virtual storefront where normal buyers browse and purchase items. *Example: Buying clothes, books, or any products directly from Amazon.*
- **Business-to-Business (B2B):** Businesses sell goods, raw materials, or services to other businesses online. Companies use digital networks to trade supplies in bulk, manage inventory, and coordinate procurement purchasing with partner firms. It represents the largest type of e-commerce by dollar volume. *Example: A smartphone maker purchasing microchips in bulk from an electronic components marketplace.*
- **Consumer-to-Consumer (C2C):** Individual consumers sell goods or services directly to other consumers using an online third-party platform. A digital platform acts as a mediator or auction host helping regular people list their used or homemade items so other individuals can buy them. *Example: Selling a used laptop on eBay, renting a room on Airbnb.*
- **Mobile E-Commerce (M-Commerce):** Commercial transactions conducted over wireless networks using handheld mobile devices. Instead of using a desktop computer, consumers use mobile apps or mobile browsers on smartphones to shop, book trips, and pay for services. *Example: Ordering takeout food, booking a ride, or paying bills.*
- **Social E-Commerce:** Transactions enabled and driven by social media networks and online social relationships. Platforms integrate shopping tools (like 'Buy' buttons or marketplace sections) that allow users to buy items directly from feeds. *Example: Purchasing a product directly through Facebook Marketplace.*
- **Local E-Commerce:** E-Commerce focused on engaging consumers based on their current location. Online services use location data to connect nearby shoppers with local physical stores through digital deals and local pickups. *Example: Buying local restaurant discount vouchers.*

5. Key Drivers and Advantages of E-Commerce

Key Drivers of E-Commerce

- **Digital Convergence:** The interconnection of modern digital devices. This enables various hardware and software platforms to communicate, share data, and process transactions seamlessly.
- **Ubiquitous Access ("Anytime, Anywhere, Anyone"):** 24/7 market availability. This removes the limitations of physical store hours and geographic boundaries, allowing transactions to occur on-demand from any connected device.
- **Organizational Restructuring:** Flattened corporate hierarchies and operational decentralization. This empowers frontline staff with digital tools to handle complex tasks and decisions that previously required intermediate management oversight.
- **Cost and Margin Pressures:** Intensified global market competition. The widespread availability of competing products forces companies to adopt e-commerce channels to lower operational overhead, automate workflows, and protect shrinking margins.
- **Demand for Mass Customization:** A shift toward consumer-tailored offerings at scale. Modern consumers expect personalized goods and experiences, requiring businesses to deploy agile, digitally managed supply chains and interactive storefronts.

Key Advantages of E-Commerce

- **Lower Overhead Costs:** Avoids expenses tied to physical retail spaces, such as store rent, heavy building insurance, and large on-site sales teams.
- **24/7 Availability:** Allows customer purchases around the clock from any device, removing physical store hours and travel constraints.
- **Equal Market Reach:** Provides small merchants and startups direct access to a worldwide audience, rivaling the visibility of large corporations.
- **Better Margins & Cheaper Transactions:** Eliminates middlemen (disintermediation) and automates paper-based order processing, lowering transaction fees.
- **Convenience & Comparison Shopping:** Enables consumers to evaluate prices, features, and reviews instantly across multiple stores using search engines and shopbots.
- **Self-Service Support:** Lets shoppers manage account details, track packages, and resolve issues independently, reducing labor costs for merchants.
- **Mass Customization:** Leverages customer purchase profiles and behavioral data to personalize recommendations.
- **Enterprise Collaboration:** Connects suppliers, manufacturers, and distributors through digital networks to streamline supply chains and enable just-in-time replenishment.

6. Elements of a Business Model

Core Concept: A business model is a planned set of business processes and activities designed to generate a profit within an identified market.

Business Model vs. Business Plan: A business model serves as the central operational blueprint for generating revenue and creating value. A business plan is a formal, written document outlining the business model in detail, including execution schedules, budgets, market analysis, and the competitive environment.

Role in E-Commerce: An e-commerce business model explicitly leverages the unique capabilities of digital technologies—namely the Internet, the Web, and mobile applications—to deliver its value proposition.

The Eight Key Elements of a Business Model

Every viable business model must answer eight fundamental questions:

1. Value Proposition

- **Central Question:** "Why should the customer buy from you?"
- **Definition:** Outlines how a company's product or service directly fulfills customer needs and why a buyer will choose it over alternative options.
- **Key Advantages in E-Commerce:** Personalization and Customization, Reduction of Search and Discovery Costs, and Convenience (managing transaction execution and direct product delivery).

2. Revenue Model

- **Central Question:** "How will the business earn money?"
- **Definition:** Also termed the financial model, this outlines the exact mechanism through which the company generates revenue, captures profits, and produces a superior return on invested capital.

3. Market Opportunity

- **Central Question:** "What marketplace do you intend to serve, and what is its size?"
- **Definition:** Refers to the total revenue potential within a firm's target marketplace (an online area of commercial exchange unconstrained by physical or temporal boundaries).

4. Competitive Environment

- **Central Question:** "Who else occupies your intended marketplace?"
- **Definition:** The existing landscape of competitors selling similar products or operating within the same marketplace. Analyzes the number of active competitors, market share, profitability, and direct vs. indirect competitors.

5. Competitive Advantage

- **Central Question:** "What special advantages does the business bring to the marketplace?"
- **Definition:** Achieved when a firm can produce a superior product or bring goods to market at a lower price point than competitors.
- **Key Sources:** First-Mover Advantage, Asymmetry (access to unique knowledge, patents, or financial resources), and Leverage (applying existing operational advantages to adjacent, newly targeted markets).

6. Market Strategy

- **Central Question:** "How do you plan to promote your products or services to attract your target audience?"
- **Definition:** The systematic plan detailing how the firm will enter the target market, gain visibility, and convert prospects into customers. Relies on digital marketing tactics such as search engine marketing, social media campaigns, and affiliate networks.

7. Organizational Development

- **Central Question:** "What types of organizational structures within the business are necessary to carry out the business plan?"
- **Definition:** The structural roadmap defining the company's internal divisions, job roles, reporting hierarchies, and business procedures.

8. Management Team

- **Central Question:** "What kinds of experiences and background are important for the company's leaders to have?"
- **Definition:** The individuals directly responsible for executing the operations. A competent team provides credibility to outside investors and possesses deep market-specific knowledge and agility.

7. E-Commerce Revenue Models

E-commerce businesses primarily rely on five major methods to generate revenue:

- **Advertising Revenue Model:** A website provides valuable content/services to build an audience, then sells advertising space to third-party merchants. Key metric: *Stickiness* (capacity to keep users on-site). Examples: Meta, Yahoo.
- **Subscription Revenue Model:** Customers pay an ongoing, recurring fee for access to premium content or ongoing deliveries. *Freemium Variant:* Basic functionality is free; advanced features require a paid subscription. Examples: Netflix, Spotify.
- **Transaction Fee Revenue Model:** The firm receives a commission or percentage for facilitating or executing an exchange between buyers and sellers. Examples: eBay, E*Trade.
- **Sales Revenue Model:** Companies generate revenue directly through the sale of physical merchandise, digital content, or discrete services to customers. Examples: Amazon, Birchbox.
- **Affiliate Revenue Model:** A site steers its visitors to partner merchants via referral links and collects a commission on any resulting sales or lead conversions.

8. Major B2C Business Models

E-commerce business models are primarily categorized by market relationship (e.g., B2C, B2B). It is important to distinguish the business model (economic architecture) from the technology platform (desktop, m-commerce) through which it operates.

Portal

- **Characteristics:** Offers integrated search capabilities alongside a consolidated package of content, news, communication tools, and entertainment to act as a general gateway to the Web.
- **Revenue Streams:** Heavily dependent on display/search advertising, alongside subscriptions and transaction referral fees.
- **Types:** General portals (Yahoo, MSN) vs. specialized vertical portals ("vortals").

E-tailer (Online Retailer)

- **Characteristics:** An online merchant that sells physical or digital products directly to consumers.
- **Primary Variations:**
 - *Virtual Merchants:* Pure-play online retailers with no physical storefronts (e.g., Amazon).
 - *Omnichannel / Bricks-and-Clicks:* Established physical retailers that also run robust e-commerce operations (e.g., Walmart).
 - *Manufacturer-Direct (D2C):* Producers selling goods directly to end users, cutting out middlemen (e.g., Dell).
 - *Catalog Merchants:* Retail operations transitioned from traditional mail-order catalogs to online web channels.
- **Revenue Model:** Sales revenue.

Content Provider

- **Characteristics:** Focuses on the production, distribution, and syndication of intellectual property and digital content, including news, video, music, photography, and text.
- **Revenue Streams:** Subscriptions, pay-per-view/download, advertising, and affiliate referral links.

Transaction Broker

- **Characteristics:** Processes transactions that are traditionally executed in person, via telephone, or through independent agents.
- **Core Value:** Reduces transaction friction and simplifies booking/trading by providing an automated online interface.
- **Revenue Model:** Transaction fees and commissions.
- **Examples:** Online travel agencies (Expedia), online financial brokers (E*Trade).

Market Creator

- **Characteristics:** Establishes a digital platform where buyers and sellers can discover one another, list items, browse offerings, negotiate terms, and determine prices.
- **Distinction:** Unlike transaction brokers, market creators do not act as agents for the parties; they provide the digital environment and infrastructure for buyers and sellers to interact directly.
- **Revenue Model:** Listing fees, access fees, and transaction commissions.
- **Examples:** eBay, Uber, Airbnb.

Service Provider

- **Characteristics:** Sells or provides online services, professional expertise, cloud software capabilities, or data processing rather than tangible products.
- **Revenue Streams:** Service fees, ongoing subscriptions, freemium upgrades, and ad placements.

Community Provider (Social Network)

- **Characteristics:** Cultivates a digital space where users with shared interests, backgrounds, or social circles can communicate, publish user-generated content (UGC), and interact.
- **Modern Expansion:** Modern community providers increasingly integrate commercial tools, enabling peer-to-peer buying and selling directly within the social ecosystem.
- **Revenue Streams:** Hybrid model combining high-volume targeted advertising, premium subscriptions, transaction fees, and affiliate referral commissions.
- **Examples:** Meta (Facebook), LinkedIn, Reddit.

9. Raising Capital & Crowdfunding

Raising Capital

Raising capital refers to how new e-commerce startups get the money they need to launch and operate before they become profitable.

- **Seed Capital / Bootstrapping:** Using personal savings, credit cards, or borrowing from friends and family to build the first prototype or website.
- **Incubators & Accelerators:** Organizations (like Y Combinator or TechStars) that provide small seed funding, office space, and business mentorship in exchange for a small percentage of equity.
- **Angel Investors:** Wealthy individuals who invest their own personal money in early-stage startups (usually during early rounds like Seed or Series A).
- **Venture Capital (VC) Firms:** Professional investment firms that manage pooled funds from institutions and invest large sums into high-growth, scalable startups.
- **Commercial Bank Loans:** Traditional debt financing, though less common for early tech startups due to lack of physical collateral.

Crowdfunding

Crowdfunding uses the Internet and social platforms to collect small amounts of money from a large pool of ordinary people (the "crowd") rather than relying solely on traditional banks or venture capitalists.

Primary Types of Crowdfunding:

1. **Donation / Donor-Based:** People give money to support a cause, personal project, or disaster relief with no financial return (e.g., GoFundMe).
2. **Rewards-Based:** Contributors pledge money in exchange for a future perk, tiered reward, or early copy of the product once manufactured (e.g., Kickstarter, Indiegogo).
3. **Equity Crowdfunding:** Backers invest money in exchange for actual ownership (shares or equity) in the startup. (e.g., Wefunder, StartEngine, Republic).

Key Benefits of Crowdfunding:

- **Market Validation:** Proves whether real customers actually want the product before mass production begins.
- **Wider Reach & Inclusivity:** Gives female and minority founders, as well as businesses outside traditional tech hubs, direct access to capital.

10. E-Commerce Business Model Architectures

E-commerce interactions can be visualized through various architectural models describing the flow of goods, orders, and money between entities.

Business-to-Business (B2B)

A website following the B2B model sells its products to an intermediate buyer who then sells the product to the final customer. For example, a wholesaler places an order on a company's website, receives the consignment, and sells the end product to the final customer at retail outlets.

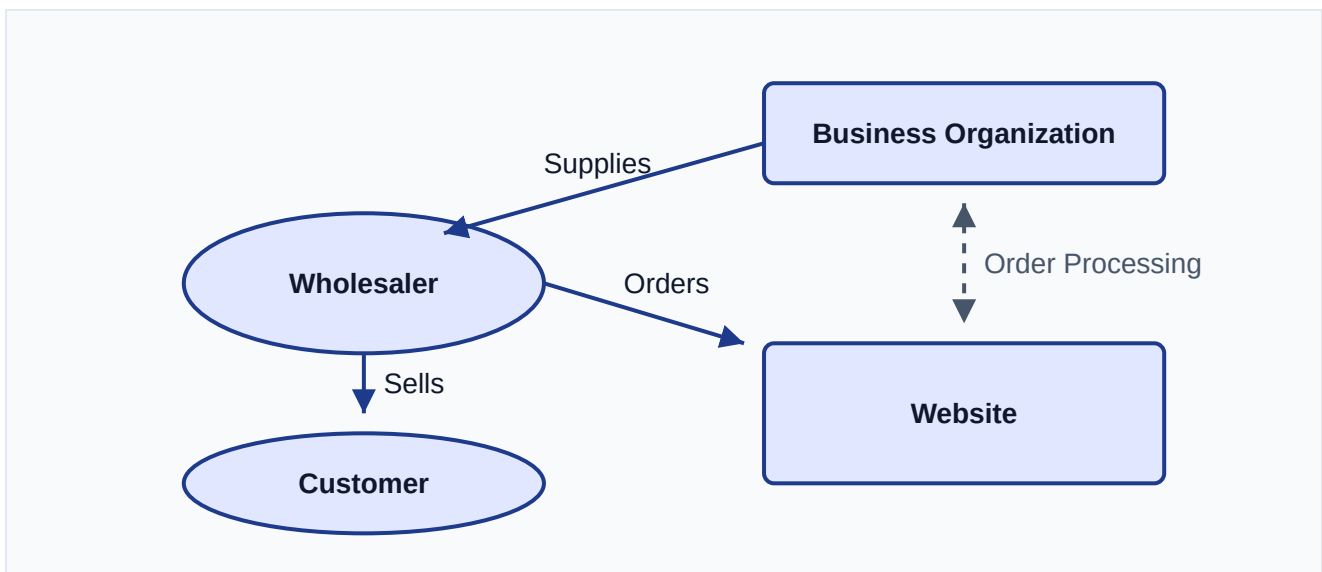
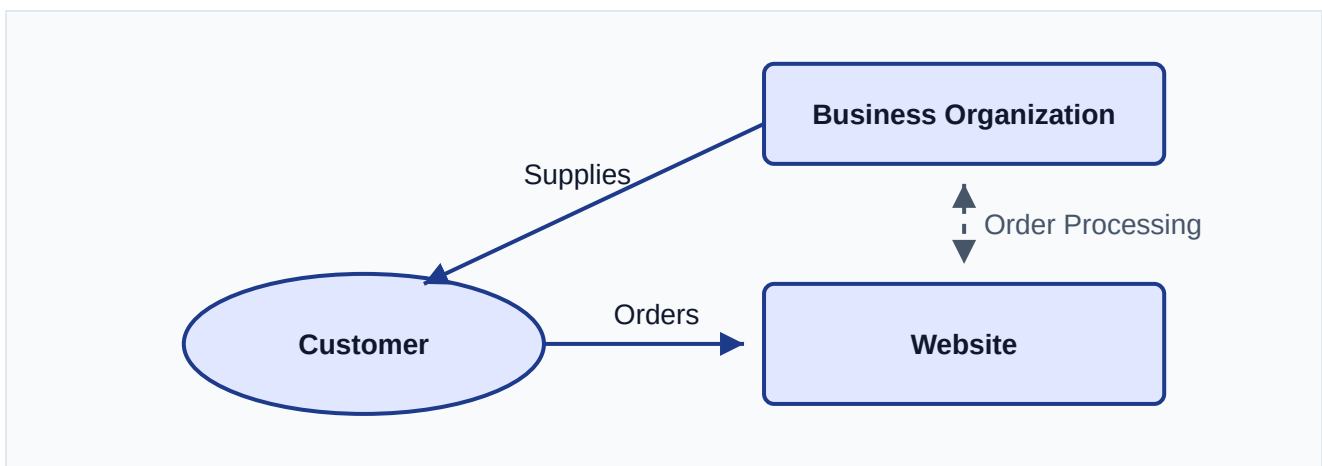


Figure 1: Business-to-Business (B2B) Interaction Model

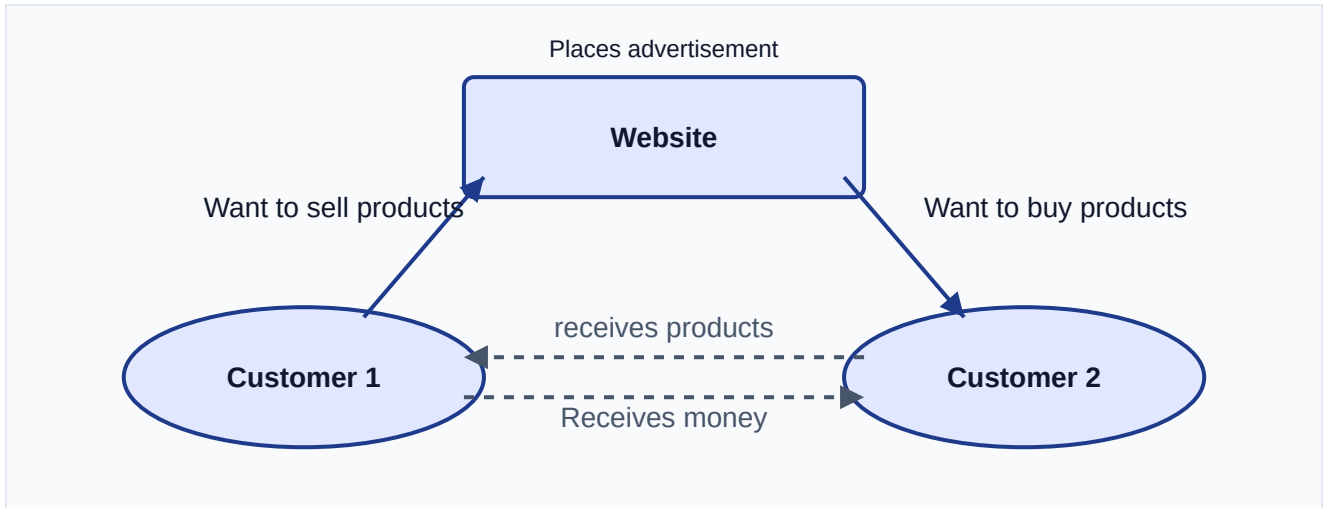
Business-to-Consumer (B2C)

A website following the B2C model sells its products directly to a customer. A customer views products on the website, chooses a product, and places an order. The website notifies the business organization, which then dispatches the goods to the customer.



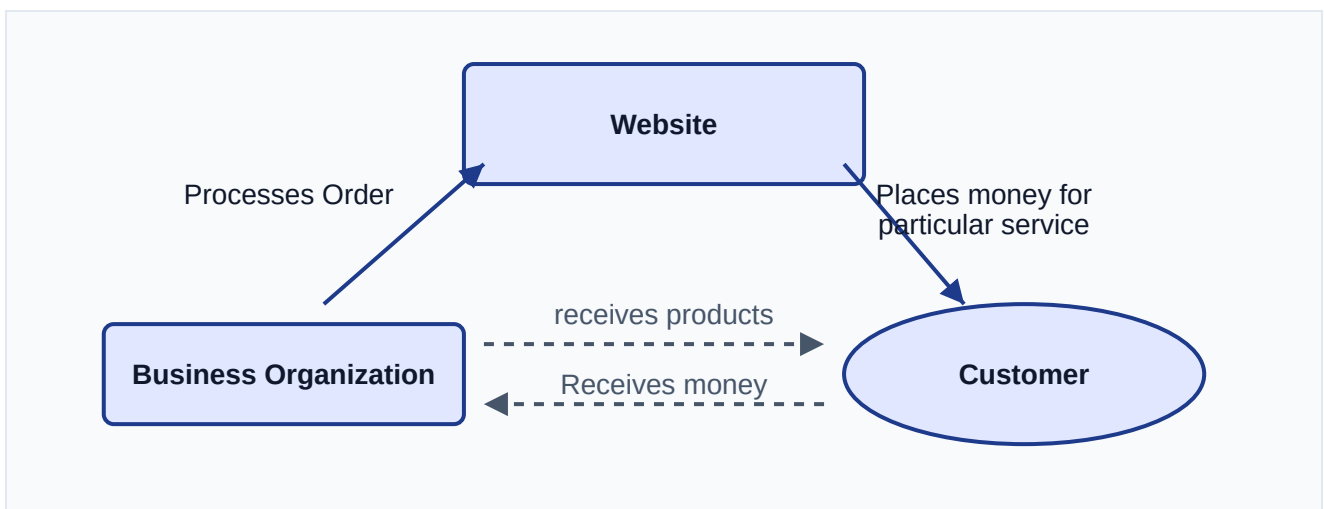
Consumer-to-Consumer (C2C)

A website following the C2C model helps consumers sell their assets (property, cars) or rent a room by publishing information online. Another consumer views the post/advertisement and buys the product from the first consumer.



Consumer-to-Business (C2B)

In this model, a consumer places an estimate of the amount they want to spend for a service on a website showing multiple business organizations. A business organization that fulfills the consumer's requirement within the specified budget approaches the customer to provide the service.



Government E-Commerce Models

- **Business-to-Government (B2G):** A variant of B2B. These websites are used by governments to trade and exchange information with business organizations. Such websites are accredited by the government and provide a medium for businesses to submit application forms to the government. (*Business Organization → Website → Government*)
- **Government-to-Business (G2B):** Governments use G2B websites to approach business organizations. These websites support auctions, tenders, and application submission functionalities. (*Government → Website → Business Organization*)
- **Government-to-Citizen (G2C):** Governments use G2C websites to approach citizens in general. They support auctions of vehicles or machinery, and provide services like registration for birth, marriage, or death certificates. The main objective is to reduce the average time for fulfilling citizens' requests for government services. (*Government → Website → Citizen*)

11. Internet Technologies, Protocols & SEO

Key Protocols Accessible on the Web

- **E-mail:** The protocol for e-mail is Simple Mail Transport Protocol, or *SMTP*.
- **HTTP:** *Hypertext Transfer Protocol* makes possible the transmission of hypertext over networks.
- **VoIP:** *Voice over Internet Protocol* makes it possible to place a telephone call over the Web.
- **FTP:** *File Transfer Protocol* is a standard protocol that allows you to copy files from computer to computer.
- **Telnet:** A basic Internet protocol that allows you to log on to a computer and access files from a remote location as if they were local files.
- **BBS:** *Bulletin Board Systems* is a computer-based meeting and announcement system allowing local people to exchange information. It generally has a simple interface to the Internet for users to access services like e-mail and NetNews.

Search Engines & SEO

The Search Engine: A search process begins with a search engine—a Web site or a database, along with the tools to generate that database and search its contents for "keywords" that describe what you're looking for.

Tips for Search Engine Optimization (SEO)

- General keywords are nowhere as good as specific keyword phrases.
- Check the Web site of the competition for ideas.
- Think of what visitors would search for in the page you're optimizing.
- Include the most important keyword phrases in heading tags on your page.
- Finalize the list of keyword phrases for the pages you optimize.
- The title tag of your page is the most important factor to consider.
- Visitors as well as search engines read your pages by looking at keywords to see what you have to offer.